ALPACA AI TRADING AGENTS HACKATHON · AUG 28 – SEP 4, 2026

Spread Sentinel

An autonomous SPY put-credit-spread agent on Alpaca. One 30-year-tested weekly rule, sized by Claude through the Alpaca MCP server, with every decision written to a journal.

Alpaca Trading API | Alpaca MCP Server | Options · multi-leg | Claude | Node.js

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github.com/Hinderager/spread-sentinel

THE IDEA

Don't ask the AI to predict the market.

Most "AI trading agents" ask a model which way price goes next. Spread Sentinel doesn't.

It runs **one boring, well-tested trade** — sell a defined-risk SPY put spread ~3% below the market each week and let it expire — and uses AI for the part a rule can't do:

- **whether** this week is a week to trade,
- **how far** out of the money (3% or 4%),
- **how big** (70% or 50% of the account at risk).

THE RULE

Every week: sell a SPY put spread with the short strike **~3% below spot**, **$5 wide**, expiring at the end of the week. Hold to expiry. Buy it back for pennies on the last morning. **No stop-loss.**

THE EVIDENCE · 1,751 WEEKS, JAN 1993 – AUG 2026

Built on Alpaca's own historical options feed.

93–98%

of weeks are winners at 3% OTM, hold to expiry

+1.7%

mean weekly return on capital at risk over 30 years (+4.3% in 2023–26)

>3%

the only way to lose: SPY falls more than ~3% by expiry

Real last-trade prices for both legs for every week since Feb 2024 (~1,000 expired SPY contracts pulled from Alpaca), a VIX-calibrated model before that, real SPY settlement throughout.

Tested and **rejected by the same data**: stops on the short-strike touch, take-profit exits, iron condors, single-name underlyings, sitting out payroll weeks.

RISK GATES · WHY THERE IS NO STOP-LOSS

The stop that looks safe destroys the edge.

EXIT RULE (36 MONTHS)	RESULT
Hold to expiry, long strike is the only stop	+3,873%
Stop out when the short strike is touched	+11%
Take-profit / stop-multiple exits	worse than holding

60% of short-strike touches recover to a full win. A touch-stop converts the most common scare into a realised loss.

GATES THAT SURVIVE THE DATA

- **Defined risk by construction** — the long strike caps the loss; sizing is computed from that cap, never from margin.
- **Never open at judging** — buy-back at 10:30 ET on expiry day.
- **Never in two trades** — the loop refuses a new entry while one is open.
- **Credit floor** — no trade if the spread pays too little.
- **Regime abort** — index down >1.5% on the day or VIX >30 → skip.

The gate before every entry.

The agent builds the trade from live quotes, then hands it to **Claude running headless** with the official **Alpaca MCP server** and web search. Claude:

- sanity-checks the chain through the MCP tools — `get_option_snapshot`, `get_option_chain`, `get_stock_snapshot` — for stale marks, wide markets, odd skew;
- reads the calendar and the tape for the expiry window (FOMC, CPI, index-moving earnings, shock language);
- returns a JSON verdict a judge can read.

THE VERDICT

```
{ action: "enter" | "skip",
  otm: 0.03 | 0.04,
  riskPct: 0.70 | 0.50,
  reason: "…",
  headlines: [...] }
```

Rules of thumb from the backtest: calm tape & IV < 22% → 3% OTM, full size. IV 22–30%, FOMC/CPI in the window, or NVDA/MSFT/AAPL earnings → 4% OTM. Thin credit → half size. Payroll weeks are deliberately *not* a widen trigger.

ALPACA INFRASTRUCTURE

Everything runs on Alpaca's stack.

TRADING API

Account, positions, `mleg` limit orders (negative limit price = net credit), PATCH to work the price for 20 minutes, clock for market hours.

MARKET DATA API

Stock snapshot for spot; option snapshots for bid/ask, greeks and implied vol on both legs and the ATM strike.

ALPACA MCP SERVER

`alpaca-mcp-server` via `.mcp.json` — the tools Claude uses inside the gate to independently verify the chain before saying "enter".

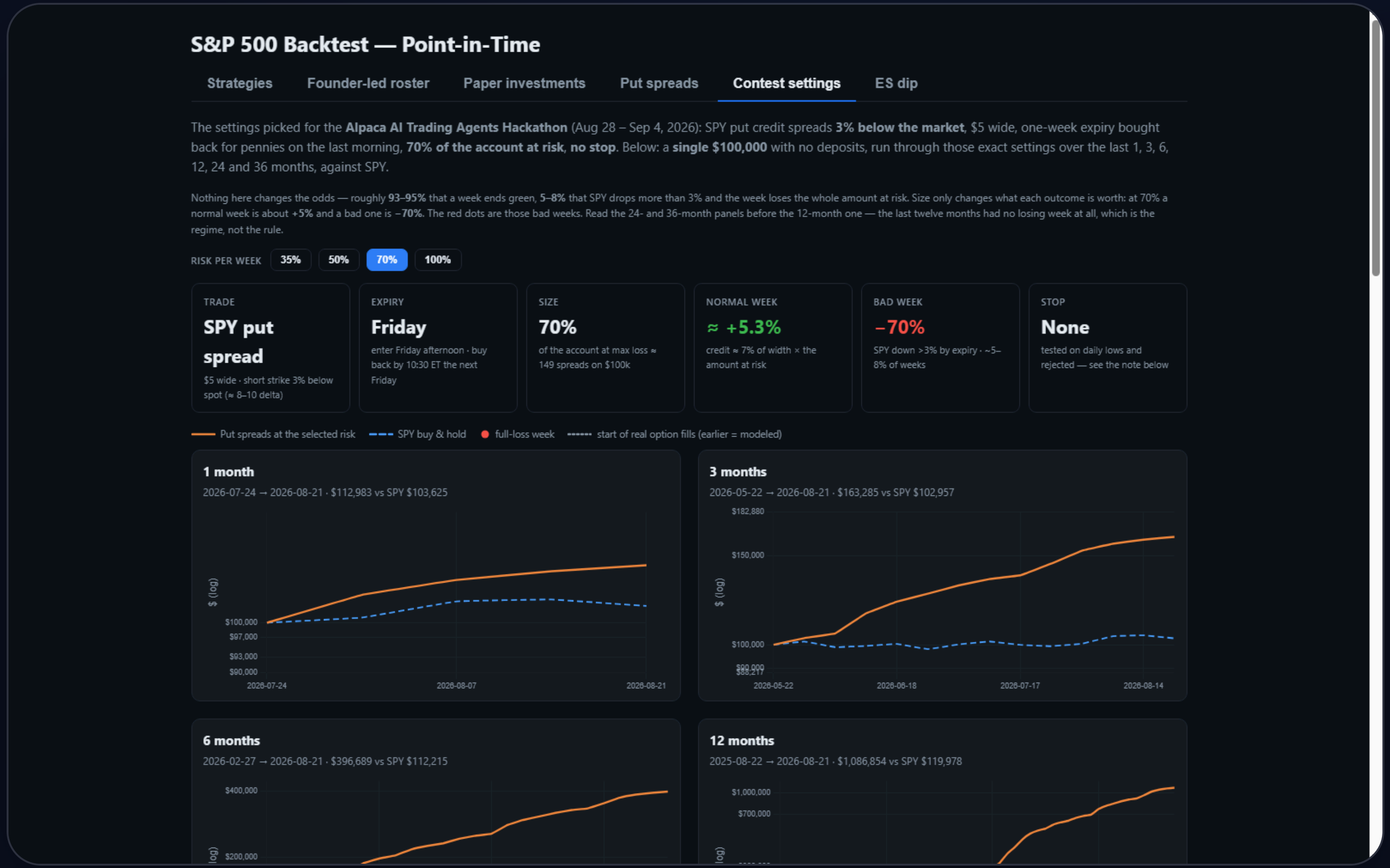
HISTORICAL OPTIONS DATA

Real last trades for ~1,000 expired SPY contracts built the weekly series the whole rule rests on.

Every decision, order, mark and buy-back — in plain text.

The agent appends to `journal/YYYY-MM-DD.md` as it goes: the gate's reasoning and full transcript, the order and fill, half-hourly marks, the buy-back and realised P/L. The public repo *is* the audit trail.

> Rehearsal week (Aug 25): sold 103 × SPY 736/731 puts for $0.17 at 50% risk with SPY at 766 and ATM IV 12.7%. Two days later the spread was worth $0.03–0.06 — up ~$1,300, SPY 4.6% above the short strike.



sp500-signal-backtest.vercel.app · Contest settings tab: the rule over 1–36 months on a single $100k, plus the live contest trades.

THE CONTEST WEEK

Fresh $100,000 account · PA3P2XKS194E

Fri

entry window 14:00–15:45 ET · SPY put spread expiring the following Friday

:30

half-hourly marks, each one published to the dashboard and the journal

10:30

ET on expiry day: buy-back near the natural so nothing is open at judging

Sizing at 70% of the account: a normal week is ≈ +5%, a bad week is −70%. The odds don't change with size — sizing is the only real lever, which is exactly why it's the AI's main decision.

Run it: `node agent.js quote` · `gate` · `enter` · `mark` · `close` · `loop`